Nikita Barinov

Machine Learning Engineer | Generative AI, LLMs & Multimodal Systems

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MIPT Master's graduate (2026). ML experience at VK, T-Bank, Sber AI and Yandex. Build production ML systems end-to-end, from experimentation and training to inference optimization, deployment and online A/B validation.

EXPERIENCE

ML Engineer | AI VK — Oct 2025 - Present

Generative AI for Advertising — July 2026 - Present

Go · Python · PyTorch · vLLM-Omni · Temporal · PostgreSQL · Kafka · S3 · Docker · Prometheus

- Built a production pipeline that transforms static banners into video ads through automated scriptwriting, image generation and video synthesis.
- Replaced template-based generation, increasing clicks by 33% and conversions by 19% while reducing CPC by 21% and CPA by 11%, with only 5% higher ad spend.
- Optimized MiniMax H3 (TI2V) inference with vLLM-Omni, achieving ~4x faster generation in benchmarks through attention backend optimization and denoising-step tuning, and reducing per-GPU memory usage by 28% with tensor parallelism.
- Implemented asynchronous generation workflows with Temporal, integrating model APIs, artifact storage and analytics.

LLM & Multimodal AI for VK Video — October 2025 - June 2026

PyTorch · vLLM · Kubernetes · Triton Inference Server · TensorRT · ONNX · Docker · DVC (S3) · YTsaurus

- Fine-tuned LLM embedding models for retrieval, improving quality on internal relevance benchmarks.
- Designed a reproducible experimentation pipeline and conducted 100+ training experiments, expanding relevance dataset coverage and diversity.
- Developed multimodal content models, improving recall by 14% at a fixed precision of 0.90.
- Distilled OpenAI o3 into a compact relevance LLM, achieving consistency recall of 0.93 and precision of 0.95 on internal evaluations.
- Deployed production models with Triton Inference Server, using TensorRT and ONNX to optimize inference.

ML Engineer | Tinkoff AI Center — Sep 2024 - Sep 2025

Python · SQL · Transformers · Sequential Models · Polars · Airflow · Docker · CatBoost

- Owned end-to-end development of Transformer-based sequential recommender for candidate generation - from research and training to production deployment; validated via online A/B test with +7% user timespent.
- Operated production ML pipelines: batch inference, retraining, monitoring, data validation.
- Automated training and deployment using Airflow + Docker.

Data Scientist Intern | Sber AI — Feb 2024 - Aug 2024

Python · PyTorch · Diffusion models · Grounded-SAM · LLaVA · Computer Vision · WebDataset · Accelerate · Distributed

- Designed and built end-to-end dataset generation pipeline (Stable Diffusion + Grounded-SAM + LLaMA + LLaVA-NeXT) for instruction-based image editing research.
- Generated millions of training samples; optimized data loading via WebDataset for high-throughput training.

ML Engineer Intern | Yandex Music — Jun 2023 - Nov 2023

Python · SQL/YQL · RecSys · Embeddings · CatBoost · Feature Engineering · Pandas · NumPy

- Integrated Transformer-based track embeddings into next-track prediction pipeline used in production recommendation system.
- Designed feature generation pipeline in Python + YQL for large-scale user interaction logs.

SKILLS

LLM & RAG: LLM, LangChain, RAG, Agentic RAG, GraphRAG, Qdrant, Vector Search, LLM Fine-Tuning, Knowledge Distillation, Information Retrieval

Deep Learning: PyTorch, Transformers, Multimodal AI, Generative AI, Image Generation, Video Synthesis, Diffusion Models, Audio ML, Whisper

ML in Production: Python, Go, vLLM, vLLM-Omni, Inference Optimization, Model Quantization, GPU Computing, Tensor Parallelism, MLflow, Triton Inference Server, TensorRT, ONNX

Recommendation Systems: Sequential Models, Transformers, Vector Search, CatBoost

Computer Vision: Stable Diffusion, Grounded-SAM, LLaVA, Object Detection

Infrastructure & MLOps: Docker, Kubernetes, FastAPI, Airflow, Temporal, Kafka, S3, Prometheus, YTsaurus, PostgreSQL, DVC[S3], Distributed Training, WebDataset

EDUCATION

MIPT - Master's, Applied Mathematics and Computer Science | Sep 2024 - Jun 2026

MIPT - Bachelor's, Applied Mathematics and Computer Science | Sep 2020 - Jun 2024

Department of Data Analysis, MIPT & Yandex School of Data Analysis | Sep 2022 - May 2024

RESEARCH PROJECTS

Garage - Generative Data Augmentation 2024

Built a Python framework for automatic object replacement in images: GroundingDINO for detection and PowerPaint for inpainting. Supports large-scale dataset augmentation with a Gradio interface.

PERSONAL PROJECTS

Audio RAG Apr 2026 - Present

Built production-ready RAG system for audio content with Triton Inference Server. Implemented speech-to-text with Whisper ASR and word-level timestamps. Integrated BGE-M3 multilingual embeddings with Qdrant vector database.

RAG Architectures Comparison Sep 2025 - Present

Implemented vanilla, graph-based and agentic RAG pipelines. Built evaluation framework for multi-hop QA. Used local LLM + embedding server.

Audio Question Answering Jan 2026 - Feb 2026

Implemented AudioLLM architecture with Whisper encoder + LLM. Built adapter-based training pipeline. Generated QA dataset from LibriSpeech.

AI News Meme Generator Jun 2025 - Jul 2025

Built automated news scraper for Russian news sources (Lenta.ru, RIA.ru). Implemented topic classification using LLaMA 3.1 with few-shot learning. Created RAG pipeline for context-aware joke generation.

ShopSpy - Price Tracker Mar 2026 - Present

Built browser extension for price tracking on Wildberries and Ozon. Implemented fake discount detector based on price history analysis. Integrated AI-powered review summary using Gemini API.

PROFESSIONAL DEVELOPMENT

YSDA Speech Processing Course | Yandex School of Data Analysis | Sep 2025 - Jan 2026

AI Agents Intensive: From LLM Queries to Agent Systems | Yandex | Apr 2026